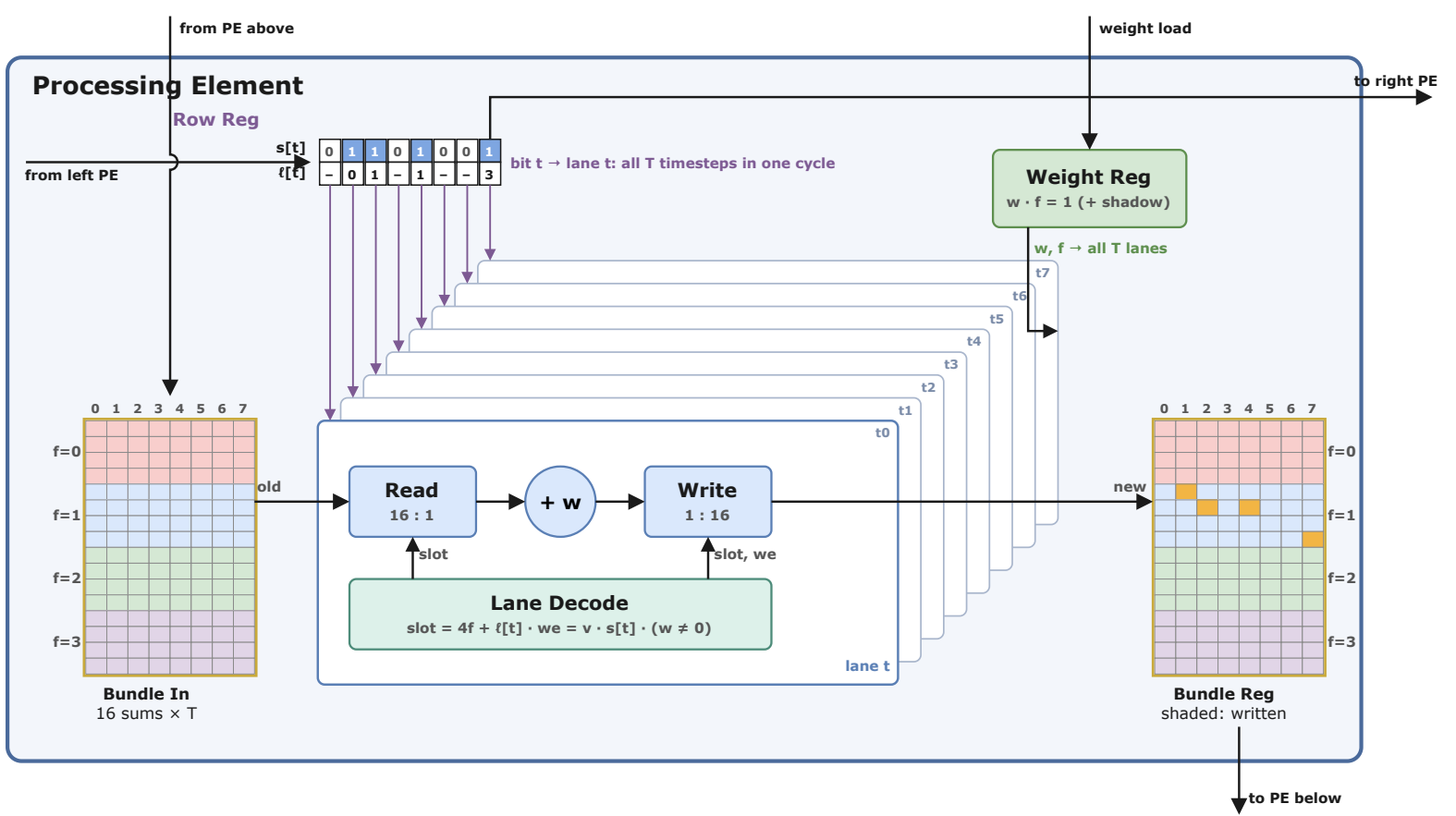

from PE above
weight load
Processing Element
to right PE
Row Reg
from left PE
s[t]
ℓ[t]
0 1 1 0 1 0 0 1
– 0 1 – 1 – – 3
bit t → lane t: all T timesteps in one cycle
Weight Reg
w · f = 1 (+ shadow)
w, f → all T lanes
t7
t6
t5
t4
t3
t2
t1
t0
0 1 2 3 4 5 6 7
f=0
f=1
f=2
f=3
old
Read
16 : 1
+ w
Write
1 : 16
slot
slot, we
Lane Decode
slot = 4f + ℓ[t] · we = v · s[t] · (w ≠ 0)
lane t
new
Bundle In
16 sums × T
Bundle Reg
shaded: written
to PE below